

CLOSED-LOOP CONTROL · 2ND-ORDER PLANT

PID Controller Simulation

Implementing and tuning a PID controller in MATLAB/Simulink to regulate a simulated second-order plant.

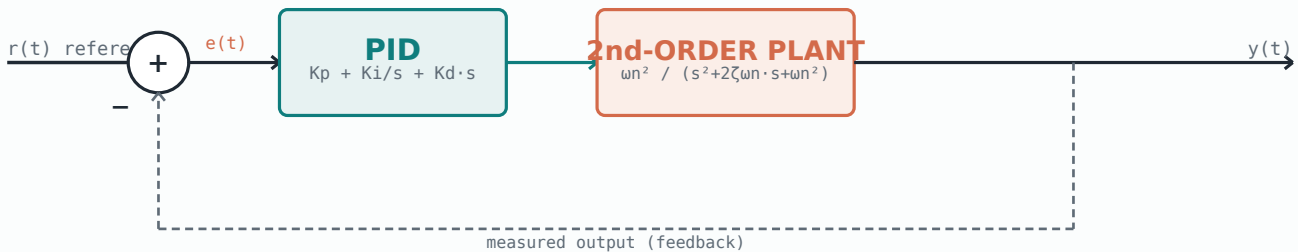


Fig. 1 – Negative-feedback loop: the controller acts on the error between reference and measured output.

Overview

This project implemented and tuned a proportional-integral-derivative (PID) controller in MATLAB/Simulink to regulate a simulated second-order plant. Second-order dynamics are the canonical model for a wide range of physical systems — mechanical mass-spring-damper assemblies, servomechanisms and RLC circuits — making this a representative study of practical closed-loop control design.

The three control terms

P Speed & stiffness $k_p \cdot \text{proportional}$ Governs the speed and stiffness of the response to error.	I Zero steady error $k_i/s \cdot \text{integral}$ Eliminates steady-state error, at the cost of added phase lag.	D Damping $k_d \cdot s \cdot \text{derivative}$ Provides anticipatory damping of overshoot and oscillation.
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Methodology

The plant was modelled in Simulink as a second-order transfer function and placed in a negative-feedback loop with a PID controller acting on the error between the reference input and the measured output. The contribution of each control term was studied systematically, building an intuition for how the gains trade off against one another.

Tuning & analysis

The three gains were tuned iteratively against quantitative time-domain objectives — principally minimising settling time and percentage overshoot in the step response while maintaining zero steady-state error. Beyond time-domain performance, the loop was analysed in the frequency domain: gain margin and phase margin were evaluated to confirm the tuned controller delivered not only fast, well-damped tracking but also a robust stability margin against model uncertainty. This dual assessment reflects standard professional practice, where a controller must be judged on robustness as well as raw performance.

Outcomes & skills demonstrated

The tuned controller achieved a fast, well-damped step response with minimal overshoot and no steady-state error, supported by healthy gain and phase margins. The project demonstrates classical control theory, transfer-function modelling, Simulink-based simulation, systematic gain tuning against defined performance criteria, and frequency-domain stability analysis — directly applicable to motor

control, power-converter regulation and robotics.